MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -1 (SET-2)

Course: B.Tech.
Session: 2024-25
Subject Name: Data Analytics
Max Marks: 20

Semester: 5
Section: A,B,C
Subject Code: BCS052
Time: 1 hr

Q . No.	1	2	3	4	5	6
CO No	CO1	CO1	CO2	CO1	CO2	CO2
Bloom's level	K4	K4	K4/K4	K2	K2/K4	K4

CO1: Describe the life cycle phases of Data Analytics through discovery, planning and building.
CO2: Understand and apply Data Analysis Techniques.

Note:
1) **This paper contains two sections (A) & (B).**
2) **All sections are compulsory.**

SECTION-A (08 Marks)

Q1.	Differentiate between analysis and reporting.	**02 Marks**
Q2.	Compare and contrast traditional analytics structure to modern analytics structure.	**03 Marks**
Q3.	Calculate the Union, Intersection and Complement of A for the following fuzzy sets A and B belonging to the universal set U: U={AA, AB, AC, AD, AE} A={(AA,0.3),(AB,0.4),(AC,0.9),(AD,0.1),(AE,0.8)} B={(AA,0.6),(AB,0.6),(AC,0.8),(AE,0.2)} **OR** Differentiate between Stochastic and Deterministic Processes. Also, explain two stochastic searching approaches.	**03 Marks**

SECTION-B (12 Marks)

Q4.	Explain various phases of the Data Analytics Life Cycle.	**04 Marks**
Q5.	Discuss various types of neural networks and their advantages and disadvantages.	**04 Marks**
Q6.	Given data= {2,3,4,5,6,7;1,5,3,6,7,8}. Compute the principal component using the PCA algorithm. **OR** Three identical boxes contain red and white balls. The first box contains 3 red and 2 white balls, the second box has 4 red and 5 white balls, and the third box has 2 red and 4 white balls. A box is chosen very randomly and a ball is drawn from it. If the ball that is drawn out is red, find the probability using Bayes theorem that the second box is chosen.	**04 Marks**

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Sessional Test-2 (SET-1)

Course: B.Tech.
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Subject Name: Data Analytics
Max Marks: 20

Semester: 5
Branch &Section: CSE (A, B, C)
Subject Code: BCS052
Time: 1 hr

Q . No.	1	2	3	4	5	6
CO No	CO5	CO5	CO3	CO3/CO4	CO4/CO5	CO3
Bloom's level	K4	K2	K4	K3/K1	K4/K3	K4

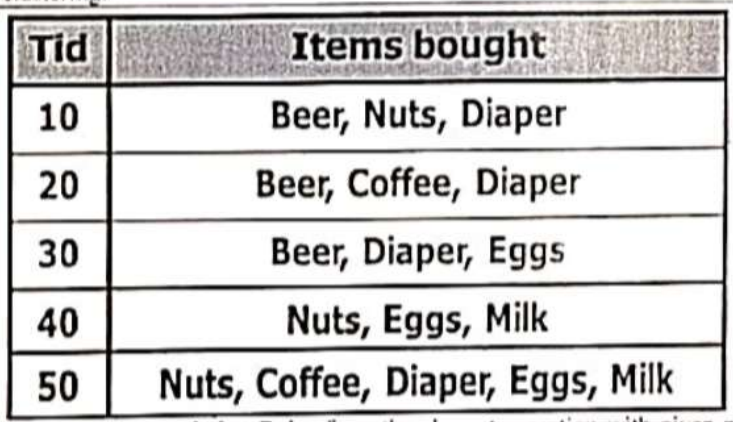
CO3 **Implement various Data streams.**
CO4 **Understand item sets, Clustering, frameworks & Visualizations.**
CO5 **Apply R tool for developing and evaluating real-time applications.**

Note:
1) **This paper contains two sections (A) & (B).**
2) **All sections are compulsory.**

SECTION-A (08 Marks)

Q1.	Differentiate between Pig and MapReduce.	02 Marks
Q2.	Explain different types of visualization techniques.	03 Marks
Q3.	Compare and Contrast DBMS and DSMS	03 Marks

SECTION-B (12 Marks)

Q4.	Explain and Apply Flajolet Martin's Algorithm on the following stream of data and identify unique elements in the stream Stream: 4, 2, 5 ,9, 1, 6, 3, 7 Hash function, h(x) = (ax + b) mod 32 a) h(x) = 3x + 1 mod 32 b) h(x) = x + 6 mod 32 **OR** Define Clustering and explain its various types. Elaborate the hierarchical methods of clustering.	04 Marks
Q5.	 Find all the Association Rules from the above transaction with given min_sup=50% and min_conf=50%. **OR** Write a function in R to check if a number is prime or not.	04 Marks
Q6.	Write note on Stock Price Prediction stating its applications.	04 Marks

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk